

APPROVED METHOD:
IES GUIDE TO LAMP SEASONING
AN AMERICAN NATIONAL STANDARD

ANSI/IES LM-54-20

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IES GUIDE TO LAMP SEASONING
AN AMERICAN NATIONAL STANDARD**

Publication of this document
has been approved by IES.
Suggestions for revisions
should be directed to IES.

**Prepared by
The IES Testing Procedures Committee**



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Foreword

This Approved Method replaces *LM-54-12, IES Guide to Lamp Seasoning*.

1.0 Introduction and Scope

1.1 Introduction

This Approved Method is intended to provide clear recommendations and requirements on procedures for seasoning of lamps to attain a state of sufficient stability for reproducible measurement of initial photometric, colorimetric, and/or electrical characteristics.

1.2 Scope

This Approved Method applies to normal and accelerated seasoning of incandescent filament lamps, cathode fluorescent lamps, and high intensity discharge (HID) lamps. Additional information on these lamp types may be found in *ANSI/IES LP-4-20, Lighting Practice: Electric Light Sources – Properties, Selection, and Specification*,¹ other ANSI/IES documents in the Lighting Measurement (LM) series, and other references listed in the Additional Reading section. Manufacturers' recommendations for seasoning should be followed for lamp types other than those listed above. Lamps intended for use as reference standards require special seasoning.²

This document does not apply to solid state lighting (SSL) devices. SSL devices are defined for this document as devices where the light is generated by electron-hole recombination at a semiconductor junction, as for example, light emitting diodes (LEDs).

2.0 Normative References

There are no normative references for this document.

3.0 Definitions

There are no definitions for this document that are not contained in ANSI/IES LS-20-1³ or ANSI C82.13:2002.⁴

4.0 Physical and Environmental Test Conditions

4.1 General Information and Cautions

It is good laboratory practice that the storage and testing of lamps should be undertaken in a relatively clean environment.

Some lamps covered in this document operate at high temperature; emit hazardous levels of optical radiation when operated, when not operated properly, or when not suitably enclosed; and/or can fail violently. Accelerated seasoning increases these hazards. Lamp manufacturers' recommendations for safety, where available, shall be followed to ensure the safety of test personnel.

4.2 Vibration

Lamps shall not be subjected to excessive vibration or shock during life testing.

4.3 Temperature

The ambient temperature is not a critical consideration for the seasoning of incandescent lamps. Bulb wall temperature and seal temperature for tungsten-halogen lamps should remain within the specifications of the manufacturer.

Ambient temperature for discharge lamps shall be controlled within the limits set by the lamp manufacturer and ballast manufacturer. These are usually between 15 °C (60 °F) and 35 °C (95 °F). Ballast case temperature at the test point specified by the manufacturer shall be less than or equal to the maximum recommended case temperature. When specified by the manufacturer, critical lamp temperatures, such as the bulb wall temperature or the control point temperature, shall be maintained. When the recommended testing

temperature range is exceeded, seasoning should be suspended. Temperatures above 40 °C (105 °F) may be deleterious to ballasts and components, and can affect lamp starting.

4.4 Airflow

Airflow is not a critical consideration for the seasoning of incandescent lamps or high intensity discharge lamps but will aid in cooling the seasoning area and the electrical components.

Airflow for fluorescent lamps, including compact fluorescent lamps, shall be minimized for proper lamp starting and operation. The lamps shall be spaced to allow airflow around each lamp.

5.0 Electrical Test Conditions

5.1 Power Supply Requirements

5.1.1 Frequency. Incandescent lamps operated under AC conditions shall use a frequency of 60 Hz. If DC or any other frequency is used, it shall be noted on the test report. For discharge lamps, the frequency shall conform to the rated frequency of the ballast.

5.1.2 Voltage Wave Shape. For AC power installations, the power supply shall have a voltage wave shape such that the total harmonic distortion does not exceed 3% of the fundamental frequency.

5.1.3 Voltage Regulation. For incandescent lamps the voltage, whether AC or DC, shall be monitored and regulated to within $\pm 0.5\%$ of the rated root-mean-square (RMS) value for the duration of the seasoning.

For discharge lamps, the ballast input voltage shall be monitored and regulated to within $\pm 10\%$ of the rated input voltage for the ballast at the specified power line frequency for the electronic control gear for the duration of the seasoning. If the seasoning time is included with the pre-burning time, the ballast input voltage shall be regulated to within $\pm 2\%$.

For integral-ballast compact fluorescent lamps, the AC voltage shall be monitored and regulated to within $\pm 10\%$ of the rated input voltage. If the seasoning time includes the pre-burning time, the ballast input voltage shall be regulated to within $\pm 2\%$.

5.1.4 Circuit Auxiliaries. For incandescent lamps, no auxiliary equipment is required.

For linear fluorescent lamps, ballasts are required. Ballasts used in seasoning shall comply with the specifications in ANSI/NEMA C82.3-2016⁵ or ANSI/NEMA C82.11-2017.⁶ If multiple lamp ballasts are used, the number of lamps required by the ballast design shall also be used.

For HID lamps, auxiliary equipment used in seasoning shall comply with the specifications in ANSI/NEMA C82.4-2017,⁷ ANSI/NEMA C82.9-2016,⁸ and ANSI/NEMA C82.17-2017.⁹ Supplemental equipment specified for use in conjunction with any auxiliary equipment shall be used.

Compact fluorescent lamps with integrated auxiliary equipment shall be seasoned as complete units. For compact fluorescent lamps without integrated auxiliary equipment, external ballasts and starters compliant with the manufacturer's guidance shall be used. The ballasts used in seasoning shall comply with the specifications in ANSI/NEMA C82.1-2004.^{6,10} If multiple-lamp ballasts are used, the number of lamps required by the ballast design shall also be used. Starters shall comply with the specifications in IEC 60081, ed. 5.¹¹

Auxiliary equipment performance should be periodically checked to ensure proper lamp performance.

5.2 Circuit Requirements

5.2.1 Wiring. All wiring of seasoning racks shall be in accordance with national, state or provincial, and local electrical codes.

For incandescent lamps and integrated compact fluorescent lamps, racks shall be wired such that the voltage drop across any number of lamp holders wired in parallel is equal to or less than 0.1% of the rated RMS

voltage. For protection of the seasoning system, where appropriate, a fuse may be located at each lamp holder position.

For fluorescent lamps, it is important to pay particular attention to the color-coding on the input wiring to the ballast. The ballast case and ground wire shall be ground, if supplied. (See **Section 5.2.3 Adjacent Ground**.) For HID lamps, the “hot” leads are connected to the center contact of the lamp holder, with the ground lead connected to the lamp holder shell.

5.2.2 Lamp Connections. For screw-base incandescent lamps, HID lamps, or integrated compact fluorescent lamps, lamp holders should be of a porcelain-type construction with sturdy threaded shells and center contacts. Lamp holders shall be rated for the specific operation under which they are used. Nickel or silver-plated lamp holders are preferred over plain brass because of their resistance to corrosion. Aluminum lamp holders are not recommended. Lamp holders used for high pressure sodium lamps shall be of a pulse type if a high-voltage pulse is used to start the lamp.¹² For non-screw-base lamps, the same principles of thermal and mechanical stability shall be applied to the lamp mounting mechanism. Inspection of lamp holder contacts should be made whenever lamps are placed on the seasoning rack. Lamp holders are to be replaced if they show signs of weak center contacts, corroded shells, or worn insulation.

For fluorescent lamps, the two diagonally opposed contact pins, across which the highest lamp voltage occurs when operating on the seasoning rack, shall be marked and referred to as the “hot pins.” Inspection of lamp holder contacts and tests for cathode-heating voltage should be made whenever lamps are placed on the seasoning rack.

5.2.3 Adjacent Ground. For fluorescent lamps (especially for rapid-start lamps), a starting aid or grounded metal strip at least 25 mm (1 in.) wide and extending essentially the full length of the lamp (cathode to cathode), should be installed behind each lamp on the seasoning rack.¹³

5.2.4 Safety Considerations. Workers in the seasoning areas shall be equipped with the appropriate protective clothing and eye protection to avoid potentially hazardous radiation, and shall be instructed as to the possible harm of excessive visible and ultraviolet radiation, which includes severe skin burns and eye damage. The high radiance of mercury and metal halide lamps in the blue spectral region is a potential eye hazard. Therefore, certain types of lamps shall not be examined visually during operation without proper protective eyewear.¹⁴ High temperature fiberglass curtains are sometimes used on seasoning racks as a protective radiation shield. While jacketed mercury vapor and metal halide lamps may individually emit only a small amount of ultraviolet energy, the collective radiation from many lamps at the close range necessary for lamp inspection and for changing lamps may be large.

Great care shall be exercised to avoid exposure to a mercury vapor or metal halide lamp with a broken or missing outer jacket. An exposure of a few seconds at close range to an unshielded arc tube operating in an open lighting unit or luminaire is sufficient to cause eye and skin damage severe enough to require emergency medical attention. A lamp with a missing or broken envelope shall be immediately turned off and removed from the seasoning rack.

6.0 Testing Procedure Requirements

6.1 Test Preparation

6.1.1 Lamp Handling. Prior to operation, lamps shall be cleaned to eliminate handling marks.

For incandescent lamps to be installed or removed from seasoning racks, the voltage to the lamp holder shall be turned off. When the lamps are to be removed from the seasoning rack, the lamps are to be handled, transported, and stored to minimize the effect of vibration and shock. Silicone or another non-conductive lubricant can be used in the lamp holder to prevent excessive vibration of the filaments. The lamps should cool to ambient temperature prior to being disturbed.

For fluorescent lamps, including compact fluorescent, to be removed for evaluation, handling, transporting, or storing, maintaining the lamps in the same orientation as during the seasoning can reduce lamp stabilization time. This practice is generally not a requirement but can be effective and is recommended with lamp types that have defined control points or cold chambers. Compact fluorescent lamps should cool for at least one hour prior to being disturbed. For HID lamps, this practice is generally followed to prevent movement of the condensed metals from their normal location during lamp operation. HID lamps should cool to ambient temperature prior to being disturbed.

6.1.2 Lamp Marking. Individual lamps should be tracked during seasoning. Lamps can be identified by markings applied directly to the lamps or by labels that can be attached to the lamps during transport and noted by seasoning rack position. The identification method selected should take into account the effect of exposure to visible light, UV and IR radiation, and heat.

6.2 Seasoning Test Procedures

6.2.1 Incandescent Filament Lamp Seasoning.

Incandescent filament lamps, regardless of application (including general lighting, automotive, aircraft, aviation, photographic, and special service) or design (miniature, tungsten-halogen, vacuum, and gas-filled), are included in this category. The incandescent filament lamp is a source in which light is produced by passing current

through a filament, usually tungsten wire, thereby heating it to incandescence. The filament is contained in a glass envelope, which is evacuated or filled with an inert gas or combination of inert gases, or with a gas mixture containing a halogen compound. The addition of a halogen to the fill gas reduces bulb blackening.

6.2.1.1 Normal Seasoning. *Normal seasoning* refers to lamp operation at rated voltage for 0.5% of rated life. Normal seasoning is generally satisfactory to achieve acceptable stability for the measurement of photometric, colorimetric, and/or electrical values. Seasoning may be performed for longer or shorter times based on documented validation that sufficient stability is achieved. For lamps that are specified by the manufacturer for performance over a range of voltage, e.g., 115 to 125 volts, the arithmetic mean voltage of the range shall be used as the rated voltage. Lamps shall be seasoned in base-up orientation, unless specified otherwise, such as for miniature and three-way lamps. The orientation used during seasoning shall be maintained throughout subsequent testing of lamp performance characteristics, and the operating orientation shall be noted in the test report.

Continuous operation shall be used for the normal seasoning of incandescent filament lamps.

6.2.1.2 Accelerated Seasoning. Accelerated seasoning is not recommended for tungsten halogen lamps and for lamps having a rated life of 100 hours or less.

Table 6-1. Approximate Accelerated Seasoning Times for Voltage-Rated Lamps and Current-Rated Lamps

Rated Life	Accelerated Seasoning Time	Operate at:
Voltage-Rated Lamps		
100 hours to 499 hours	15 minutes for each 100 rated hours	105% rated volts
500 hours to 1,000 hours	30 minutes for each 500 rated hours	115% rated volts
Over 1,000 hours	30 minutes for each 1,000 rated hours	120% rated volts
Current-Rated Lamps		
100 hours to 499 hours	15 minutes for each 100 rated hours	103% rated current (amps)
500 hours to 2,000 hours	30 minutes for each 500 rated hours	108% rated current (amps)
Over 2,000 hours	30 minutes for each 1,000 rated hours	111% rated current (amps)

Table note: An acceptable alternative method for lamps with a rated life greater than 10,000 hours is to season for 5 hours at 120% of rated volts, or at 111% of rated current.

Approximate accelerated seasoning times applicable to voltage rated lamps and current rated lamps are given in **Table 6-1**.

More-precise estimates of accelerated seasoning time in minutes (ASM) for lamps operated at rated voltage or rated current can be determined from the following equations:

For rated-voltage operation:

$$ASM = 60 \left[\frac{0.005 \times \text{rated life}}{(0.01 \times \text{percent volts})^{13}} \right],$$

and for rated-current operation:

$$ASM = 60 \left[\frac{0.005 \times \text{rated life}}{(0.01 \times \text{percent amps})^{24}} \right],$$

where:

- The numerator (0.005 × rated life) is the minimum time in hours required for normal seasoning of the lamp that is to undergo accelerated seasoning.
- *Percent volts* and *Percent amps* are expressed as the percentage value without the % sign. For example, if a voltage-rated lamp is operating at 120% of rated volts, the denominator would be expressed as: (0.01 × 120)¹³.

Continuous operation shall be used for the accelerated seasoning of incandescent filament lamps.

6.2.2 Discharge Lamps.

6.2.2.1 Discharge Lamps except T5 Fluorescent. The discharge lamp category includes cathode fluorescent, low pressure sodium, and high intensity discharge (HID) lamps (mercury vapor, metal halide, and high pressure sodium). Discharge lamps shall be seasoned and measured in the same orientation. The seasoning time shall be a minimum of 100 operating hours or as specified by the manufacturer so that measurements can reliably establish initial or rated lumens. No generally accepted method for accelerated seasoning of discharge lamps exists. No recommendation is made here for accelerated seasoning of discharge lamps. Validation of the equivalence of an accelerated seasoning method to the recommended method shall be completed and documented prior to adoption of any accelerated seasoning method.

Lamps intended to be life tested shall be cycled during seasoning. In all other cases, continuous burning of discharge lamps during seasoning is an acceptable method to shorten the time required for appropriate seasoning.

6.2.2.2 T5 Fluorescent. T5 fluorescent lamps with asymmetric cathode geometry and liquid mercury dosing shall be seasoned for 100 hours in a vertical orientation. During seasoning, the lamp cold chamber shall be at the lowest point. The location of the cold chamber shall be indicated by the manufacturer. After seasoning, any lamp movement shall be carried out carefully, with minimal vibration or shock and with the cold spot always at the lowest point. The lamp orientation shall not be changed during handling after seasoning, before measurement, or during measurement in a way that would cause the cold spot to be elevated above the horizontal plane of the lamp.

7.0 Reporting Requirements

There are no particular requirements to be reported.

ADDITIONAL READING

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Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17th Floor, New York, NY 10005-4001. Email: standards@ies.org. Fax: 212-248-5017.

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Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

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Select as applicable:

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Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:



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